

Foundations of Random forest

[Random forest]

1.0 Content Block

Preliminaries: Centered forests Centered forest is a simplified model for Breiman's original random forest, which uniformly selects an attribute among all attributes and performs splits at the center of the cell along the pre-chosen attribute. The algorithm stops when a fully binary tree of level k is built, where $k \in \mathbb{N}$ is a parameter of the algorithm.

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Disadvantages While random forests often achieve higher accuracy than a single decision tree, they sacrifice the intrinsic interpretability of decision trees. Decision trees are among a fairly small family of machine learning models that are easily interpretable along with linear models, rule-based models, and attention-based models. This interpretability is one of the main advantages of decision trees. It allows developers to confirm that the model has learned realistic information from the data and allows end-users to have trust and confidence in the decisions made by the model. For example, following the path that a decision tree takes to make its decision is quite trivial, but following the paths of tens or hundreds of trees is much harder. To achieve both performance and interpretability, some model compression techniques allow transforming a random forest into a minimal "born-again" decision tree that faithfully reproduces the same decision function. Another limitation of random forests is that if features are linearly correlated with the target, random forest may not enhance the accuracy of the base learner. Likewise in problems with multiple categorical variables.

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The above procedure describes the original bagging algorithm for trees. Random forests also include another type of bagging scheme: they use a modified tree learning algorithm that selects, at each candidate split in the learning process, a random subset of the features. This process is sometimes called "feature bagging". The reason for doing this is the correlation of the trees in an ordinary bootstrap sample: if one or a few features are very strong predictors for the response variable (target output), these features will be selected in many of the B trees, causing them to become correlated. An analysis of how bagging and random subspace projection contribute to accuracy gains under different conditions is given by Ho. Typically, for a classification problem with p features, $\sqrt{p}$ (rounded down) features are used in each split. For regression problems the inventors recommend $p/3$ (rounded down) with a minimum node size of 5 as the default. In practice, the best values for these parameters should be tuned on a case-to-case basis for every problem.

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Mean decrease in impurity feature importance This approach to feature importance for random forests considers as important the variables which decrease a lot the impurity during splitting. It is described in the book Classification and Regression Trees by Leo Breiman and is the default implementation in sci-kit learn and R. The definition is: unnormalized average importance $(x) = \frac{1}{n_T} \sum_{i=1}^n \sum_{\text{node } j \in T_i | \text{split variable } (j) = x} p_{T_i(j)} \Delta_i T_i(j)$, where

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Consistency results Assume that $Y = m(\mathbf{X}) + \varepsilon$, where ε is a centered Gaussian noise, independent of $\mathbf{X}$, with finite variance $\sigma^2 < \infty$. Moreover, $\mathbf{X}$ is uniformly distributed on $[0, 1]^d$ and m is Lipschitz. Scornet proved upper bounds on the rates of consistency for centered KeRF and uniform KeRF.

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Uniform forest Uniform forest is another simplified model for Breiman's original random forest, which uniformly selects a feature among all features and performs splits at a point uniformly drawn on the side of the cell, along the preselected feature.

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$K_{uf}(\mathbf{0}, \mathbf{x}) = \sum_{j=1}^d k_j = k_1! \dots k_d! (1/d)^k \prod_{m=1}^d (1 - |x_m| \sum_{j=0}^{k_m-1} \frac{(-\ln |x_m|)^j}{j!})$ for all $\mathbf{x} \in [0, 1]^d$.

$$K_{k_1, \dots, k_d}(\mathbf{x}) = \sum_{j=1}^d k_j! \left(\frac{1}{d} \right)^k \prod_{m=1}^d \left(1 - |x_m| \sum_{j=0}^{k_m-1} \frac{(-\ln |x_m|)^j}{j!} \right)$$
 for all $\mathbf{x} \in [0, 1]^d$.

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Unsupervised learning As part of their construction, random forest predictors naturally lead to a dissimilarity measure among observations. One can analogously define dissimilarity between unlabeled data, by training a forest to distinguish original "observed" data from suitably generated synthetic data drawn from a reference distribution. A random forest dissimilarity is attractive because it handles mixed variable types very well, is invariant to monotonic transformations of the input variables, and is robust to outlying observations. Random forest dissimilarity easily deals with a large number of semi-continuous variables due to its intrinsic variable selection; for example, the "Addcl 1" random forest dissimilarity weighs the contribution of each variable according to how dependent it is on other variables. Random forest dissimilarity has been used in a variety of applications, e.g. to find clusters of patients based on tissue marker data.

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History The general method of random decision forests was first proposed by Salzberg and Heath in 1993, with a method that used a randomized decision tree algorithm to create multiple trees and then combine them using majority voting. This idea was developed further by Ho in 1995. Ho established that forests of trees splitting with oblique hyperplanes can gain accuracy as they grow without suffering from overtraining, as long as the forests are randomly restricted to be sensitive to only selected feature dimensions. A subsequent work along the same lines concluded that other splitting methods behave similarly, as long as they are randomly forced to be insensitive to some feature dimensions. This observation that a more complex classifier (a larger forest) gets more accurate nearly monotonically is in sharp contrast to the common belief that the complexity of a classifier can only grow to a certain level of accuracy before being hurt by overfitting. The explanation of the forest method's resistance to overtraining can be found in Kleinberg's theory of stochastic discrimination. The early development of Breiman's notion of random forests was influenced by the work of Amit and Geman who introduced the idea of searching over a random subset of the available decisions when splitting a node, in the context of growing a single tree. The idea of random subspace selection from Ho was also influential in the design of random forests. This method grows a forest of trees, and introduces variation among the trees by projecting the training data into a randomly chosen subspace before fitting each tree or each node. Finally, the idea of randomized node optimization, where the decision at each node is selected by a randomized procedure, rather than a deterministic optimization was first introduced by Thomas G. Dietterich. The proper introduction of random forests was made in a paper by Leo Breiman, that has become one of the world's most cited papers. This

paper describes a method of building a forest of uncorrelated trees using a CART like procedure, combined with randomized node optimization and bagging. In addition, this paper combines several ingredients, some previously known and some novel, which form the basis of the modern practice of random forests, in particular: